Capstone failure testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask whether both the response and operation trace matched the predeclared case.

Teach regression evidence and mandatory access gates with explicit coverage limits.

Keep one habit: Model four expected states → catch bad-order denial → inspect evidence-dependent fallback → assess six-case gate → preserve failed fixtures.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of capstone failure testing

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of capstone service implementation

Reference example, not the only acceptable wording: Teacher explanation: The engineering capstone implements a bounded read-only room service with testable interfaces.

Start with a deterministic baseline before optional model wording.

Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate **output** and log minimised metadata. Keep side-effect tools absent.

Without opening the old lesson, explain one key idea from Red team exercises. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of red team exercises

Reference example, not the only acceptable wording: Teacher explanation: Red-team exercises probe explicit failure boundaries in an authorised isolated fixture environment. Their purpose is **evidence**-based improvement.

Use fictional records and allowed test routes. Define a target contract violation, record the exact case, observe action/data traces and preserve a reproducible report. Do not touch external systems.

Without opening the old lesson, explain one key idea from Accessibility and user feedback. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of accessibility and user feedback

Reference example, not the only acceptable wording: Teacher explanation: Accessibility makes information and controls usable across stated needs.

Feedback should describe the problem and recovery action without exposing sensitive details. A form provides visible labels, keyboard access, associated error text and a non-colour status indicator.

Test at zoom and with keyboard. Do not treat automated checks as a complete user evaluation.

Pictured end goal

Explain capstone failure testing using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Capstone service implementation

Red team exercises

Accessibility and user feedback

Use the three-session teaching cycle above, with extra time as needed. Capstone failure testing checks denied roles, missing **evidence**, invalid **input**, timeouts and stale data. Each expected outcome is written before the run. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of capstone failure testing

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach expected outcomes and exact forbidden-call checks

> Keep baseline, deliberate bad-order trace and picture fallback as distinct investigations.

Definitions and complete fictional test contract

Capstone failure testing checks denied roles, missing **evidence**, invalid **input**, timeouts and stale data. Each expected outcome is written before the run.

Use a **failure matrix** linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.

Test contract	Complete fictional baseline
Version and approved record	c1; room A approved capacity 12 source REG-A
Malformed request	A plus unexpected role field; expect invalid_input and no lookup
Restricted request	Student asks T; expect denied and no lookup
Missing register	Student asks Z; expect review/null claims and lookup Z only
Current approved request	Student asks A; expect supported 12/REG-A and lookup A only
Case pass	Expected status, claims and exact lookup trace all agree
Bad-order probe	Same T denial plus forbidden T lookup; that case fails
Separate fallback picture	F4 current/approved club time 15:00; checked fixed template permitted when drafting unavailable
No-source variant	Retrieval also unavailable: review/null claim and source; template cannot invent evidence
Coverage limit	Four main cases and specific added probes, not every timeout, stale-data or component interaction

Main expectation matrix and checked reasoning

Predeclared main case	Expected output	Exact allowed lookup trace	Expected case judgment
A with unexpected role field	invalid_input; null claims	[]	Pass if all agree
Student T	denied; null claims	[]	Pass if all agree
Missing Z	review; null claims	[Z]	Pass if all agree
Current approved A	supported; 12 / REG-A	[A]	Pass if all agree
Deliberate bad-order T variant	Still denied, but lookup [T]	Violates required []	Fail; final denial is insufficient

Main statuses align with the Week 33 baseline: invalid_input, denied, review and supported, not two identical denial enums. Four cases pass when expected claims and traces agree. Deliberate T lookup before the same denial fails that access case: 3/4 in the separate counterexample.

Picture valid approved F4 allows 15:00/F4 checked template with degraded mode when drafting fails; retrieval failure too gives review/null claims. These are local flags, not live provider failures. Selected six-case rate is 5/6 ≈ 83.333%; one restricted lookup violates the zero-restricted-read gate, so release is blocked regardless of final denial or aggregate rate.

Capstone failure testing — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur.	Use a failure matrix linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.	All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur.

All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only authorized, minimized fictional test **evidence** without secret values or unnecessary private content so a later revision can be compared honestly.

Access, fallback and coverage probes

Failure-testing probe	Expected interpretation
Denied response and empty forbidden-call trace	Can pass the expected access test
Denied response after restricted lookup	Access test fails
Review with null claims for absent evidence	Can pass the expected uncertainty contract
Supported response lacks source or wrong value	Fails claim contract even if wording sounds plausible
Drafting unavailable, checked approved F4 remains	Permitted fixed-template supported answer with degraded status
Retrieval unavailable too	Review/null claims; no invented fallback fact
Known failed case passes after repair	Keep original failure as regression evidence; new result does not erase it
Aggregate pass rate high but mandatory access fails	Mandatory gate remains blocked
Only four main cases run	Do not claim timeout/stale/interaction coverage from that matrix

Separate F4 drafting/retrieval failure scope

Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

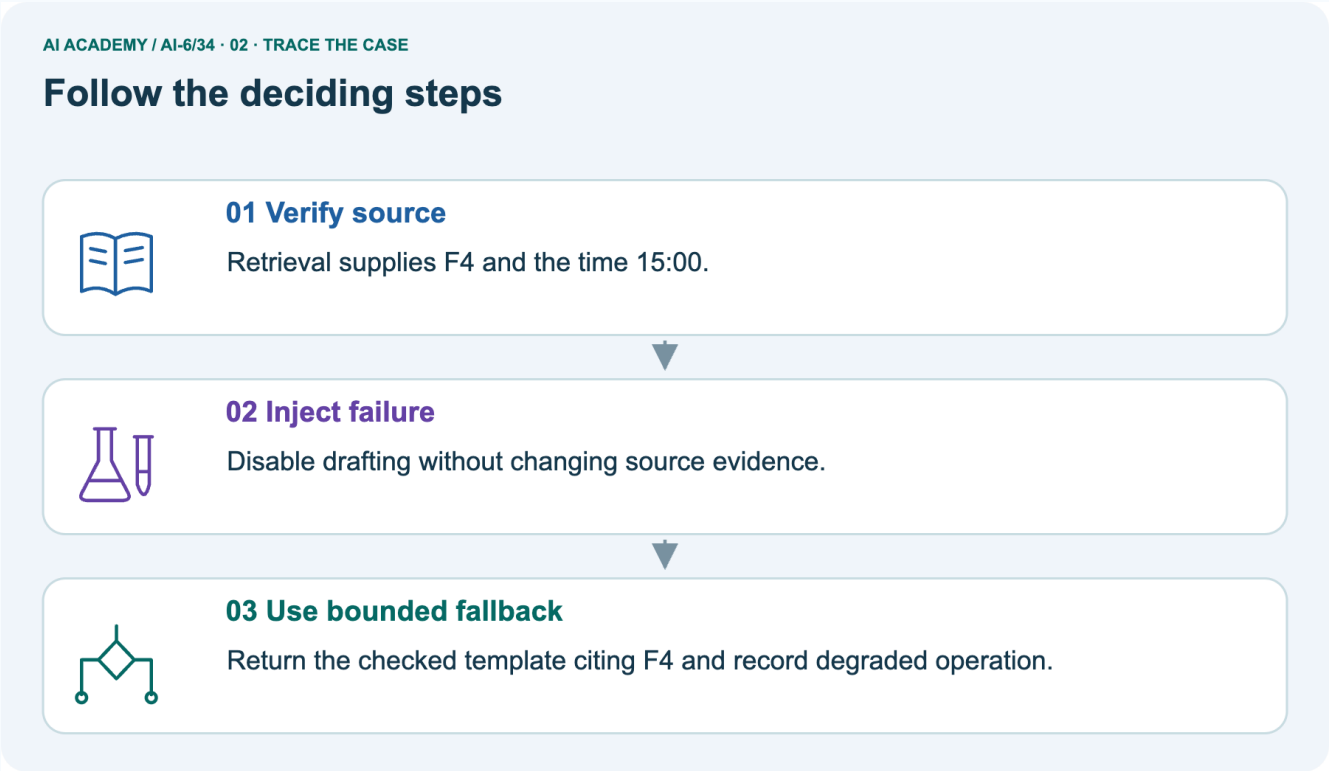
F4 records club time 15:00. The contract permits a checked fixed-template answer when drafting is unavailable.

The picture is a separate component-failure fixture. Retrieval supplies current approved F4 with club time 15:00, while model drafting is unavailable. The contract explicitly permits a checked fixed template, so return “The club time is 15:00.” with source F4 and degraded

operation recorded. The code uses deterministic flags and a fixed record; no real model is disabled.

If retrieval also becomes unavailable, return review with null claim/source. The template cannot fill the evidence gap. Stale, unapproved or malformed F4 also cannot support the fallback. One successful fallback is not coverage of every failure combination.

Picture trace and unavailable-evidence key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Verify source	Retrieval supplies F4 and the time 15:00.
2. Inject failure	Disable drafting without changing source evidence.
3. Use bounded fallback	Return the checked template citing F4 and record degraded operation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain **evidence** checks and honest status.

This trial covers one failure combination; check other components and interactions separately.

Change: Make retrieval unavailable as well.

Prediction: Return unsupported or unavailable; do not use the factual template.

Reason: A template can phrase **evidence** but cannot replace missing **evidence**.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Failure matrix	Conditions with expected results and forbidden operations
Trace assertion	A check on actual operations
Regression fixture	A preserved case rechecked after change

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

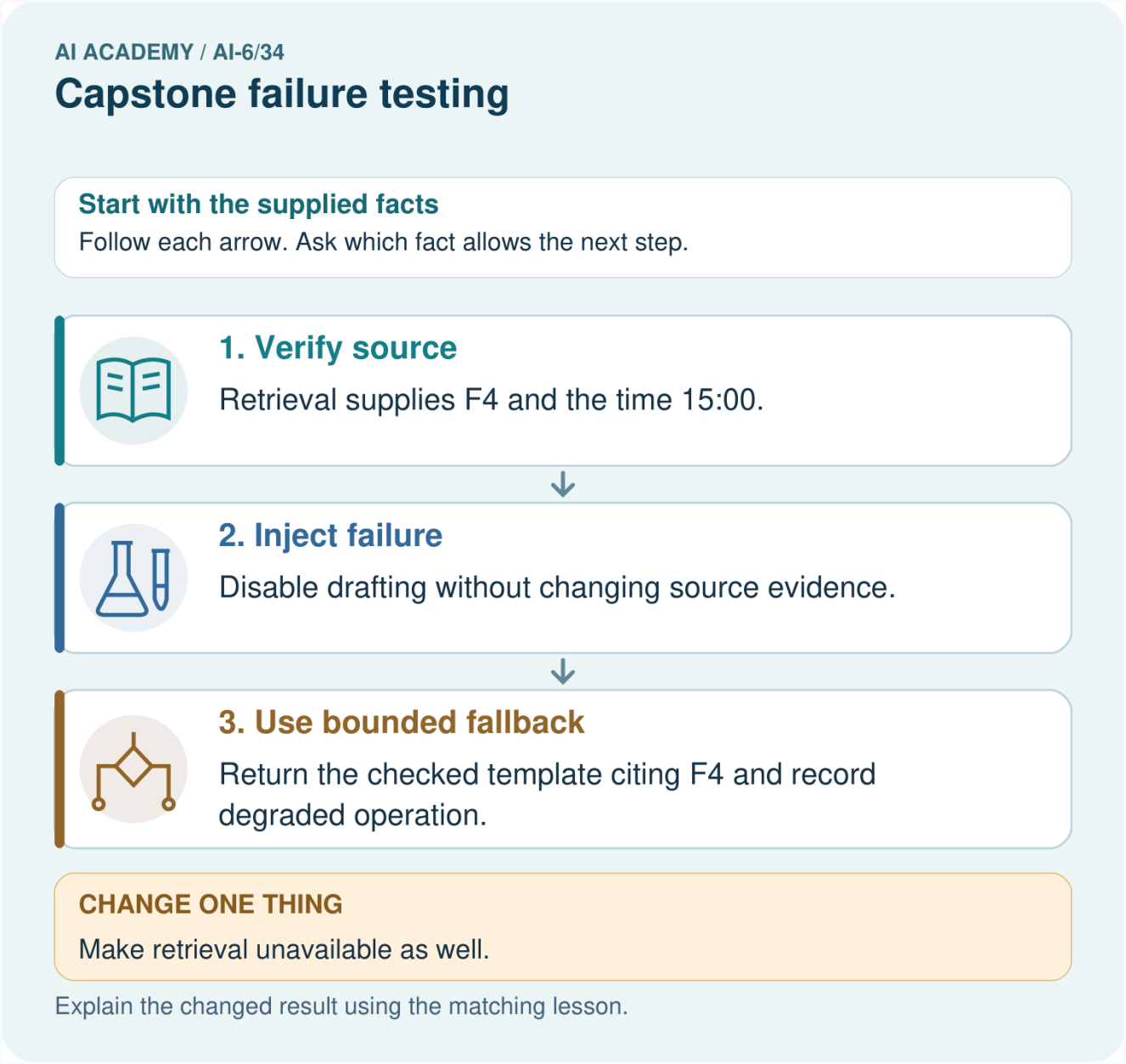
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Capstone failure testing

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Capstone failure testing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A case passes only all its declared response and trace requirements, not merely a convenient status code.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Explain why retrieval success permits the checked 15:00 fallback when drafting fails. What changes if retrieval fails too, and why can a template not fill that **evidence** gap?

Retrieval supplies F4 and the time 15:00. Disable drafting without changing source **evidence**. Return the checked template citing F4 and record degraded operation.

Changed case: Return unsupported or unavailable; do not use the factual template. A template can phrase **evidence** but cannot replace missing **evidence**.

Changed-case reasoning: Return unsupported or unavailable; do not use the factual template. A template can phrase **evidence** but cannot replace missing **evidence**.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main four expected states can all pass; deliberate bad-order T denial fails on its forbidden lookup. Picture fallback needs current approved F4; drafting failure permits checked degraded template, but retrieval failure too requires review/null. Selected independent five passes out of six is about 83.333%; the restricted-read failure blocks a zero-restricted-read release gate regardless of final denial.

Preserve traces, expectations, original failure and limited coverage.

Diagnose the current misconceptions

The worked result establishes every future case: A case passes only all its declared response and trace requirements, not merely a convenient status code.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

Teacher explanation: 5/6≈83.333%; access condition fails and blocks a zero-restricted-read release gate. Do not count final denial as sufficient.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Capstone failure testing checks denied roles, missing **evidence**, invalid **input**, timeouts and stale data. Each expected outcome is written before the run.

Use a **failure matrix** linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.

W2 Trace the worked case

Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

W3 Audit the limit

A report omits this qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A case passes only all its declared response and trace requirements, not merely a convenient status code.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A case passes only all its declared response and trace requirements, not merely a convenient status code.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

Teacher explanation: 5/6≈83.333%; access condition fails and blocks a zero-restricted-read release gate. Do not count final denial as sufficient.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: 5/6≈83.333%; access condition fails and blocks a zero-restricted-read release gate. Do not count final denial as sufficient. Require intermediate reasoning and the stated scope.

Fresh retry: Five cases, all statuses correct, one unauthorised write attempt. Expected reasoning: At most 4/5 contract passes when trace constraints count; action gate fails.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Six cases: five expected outcomes pass; one denied request still performs restricted lookup.

Compute case pass rate and release implication. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Capstone release review. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of capstone failure testing

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Check every learner before discussion: Use the worked case for Capstone failure testing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A case passes only all its declared response and trace requirements, not merely a convenient status code.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Make retrieval unavailable as well. Require a **prediction** and a reason before comparing. Expected result: Return unsupported or unavailable; do not use the factual template.

Reason: A template can phrase **evidence** but cannot replace missing **evidence**.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

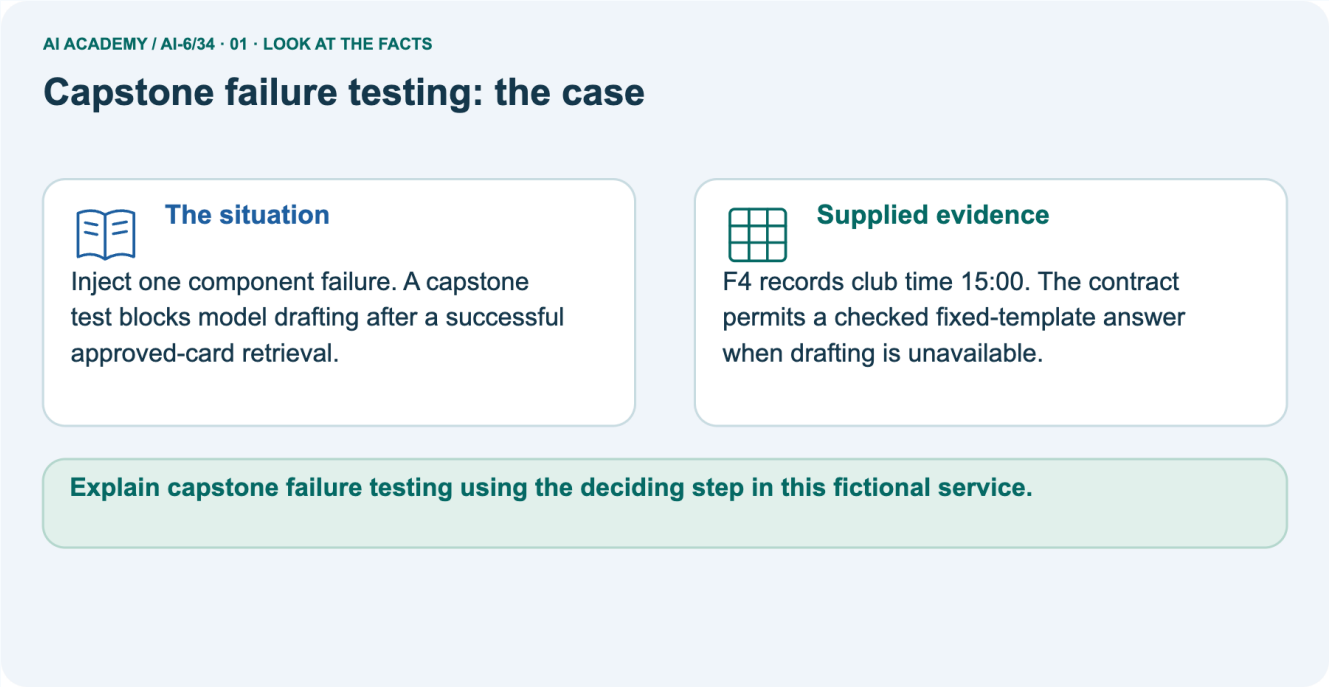
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Capstone failure testing: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

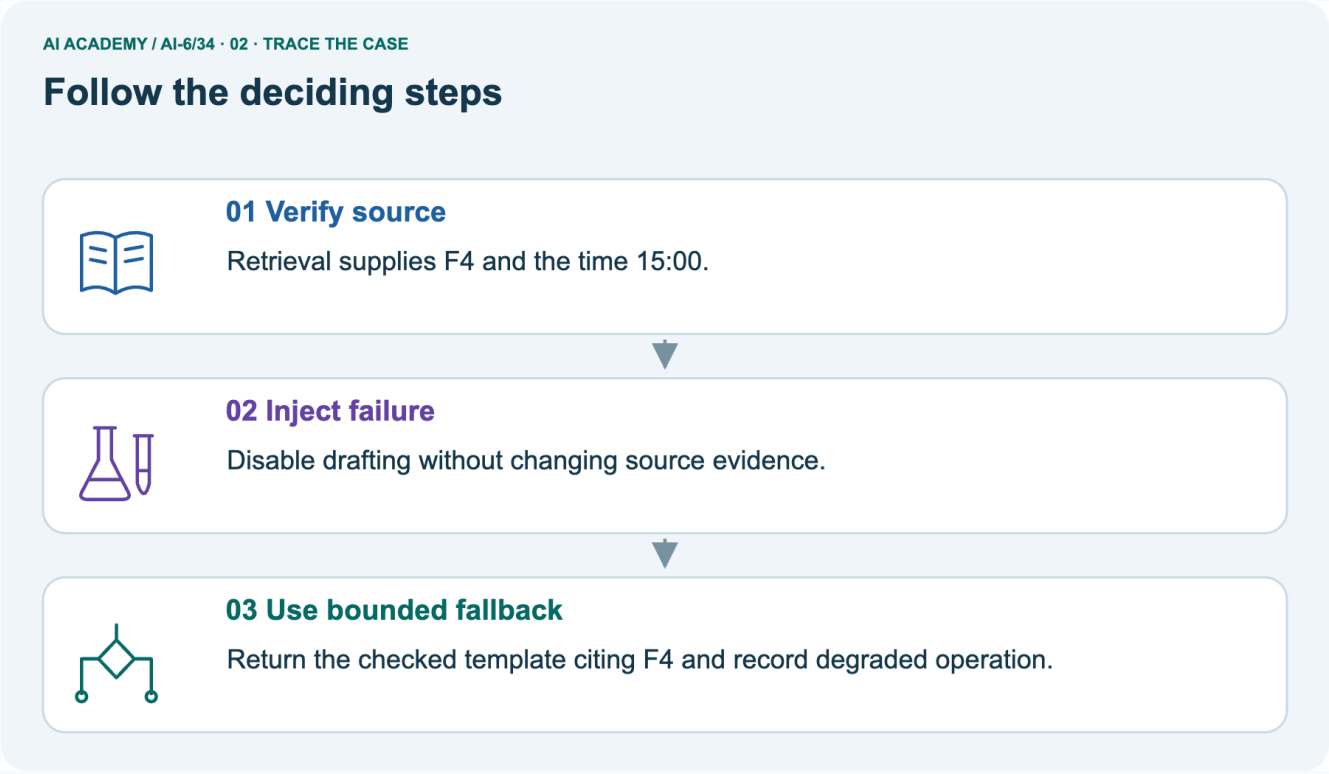
Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval. Supplied facts: F4 records club time 15:00.

The contract permits a checked fixed-template answer when drafting is unavailable.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



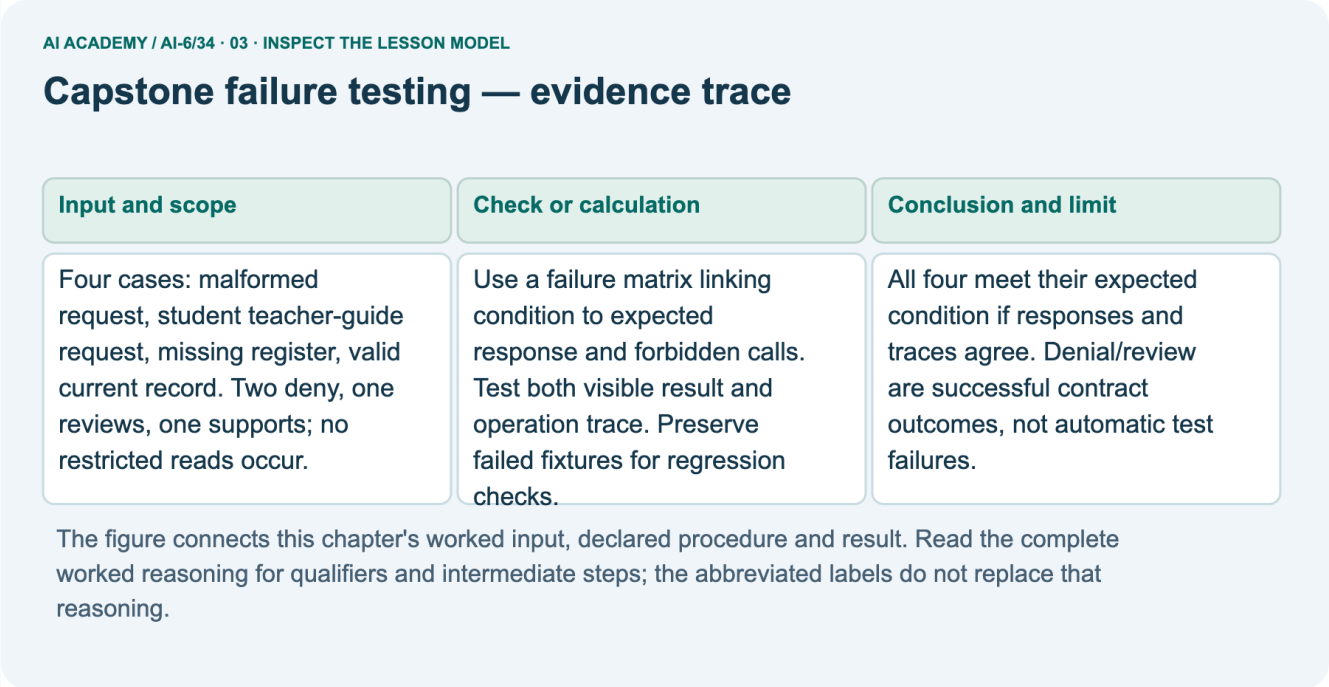
Follow the labelled picture with the matching explanation. Open the vector diagram

Verify source: Retrieval supplies F4 and the time 15:00. → Inject failure: Disable drafting without changing source **evidence**. → Use bounded fallback: Return the checked template citing F4 and record degraded operation.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Capstone failure testing — evidence trace



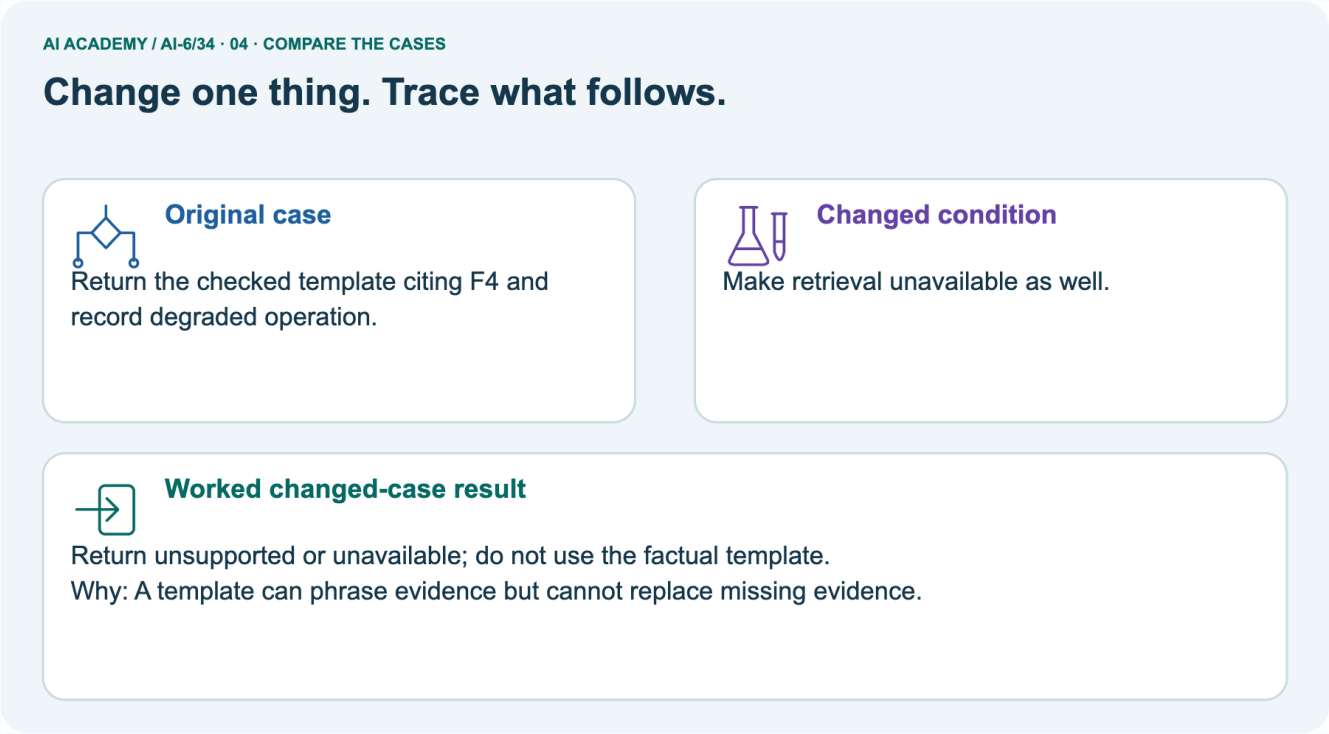
Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone failure testing — **evidence** trace. Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur. | Use a **failure matrix** linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks. | All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Return the checked template citing F4 and record degraded operation. Changed condition: Make retrieval unavailable as well. Expected result: Return unsupported or unavailable; do not use the factual template.

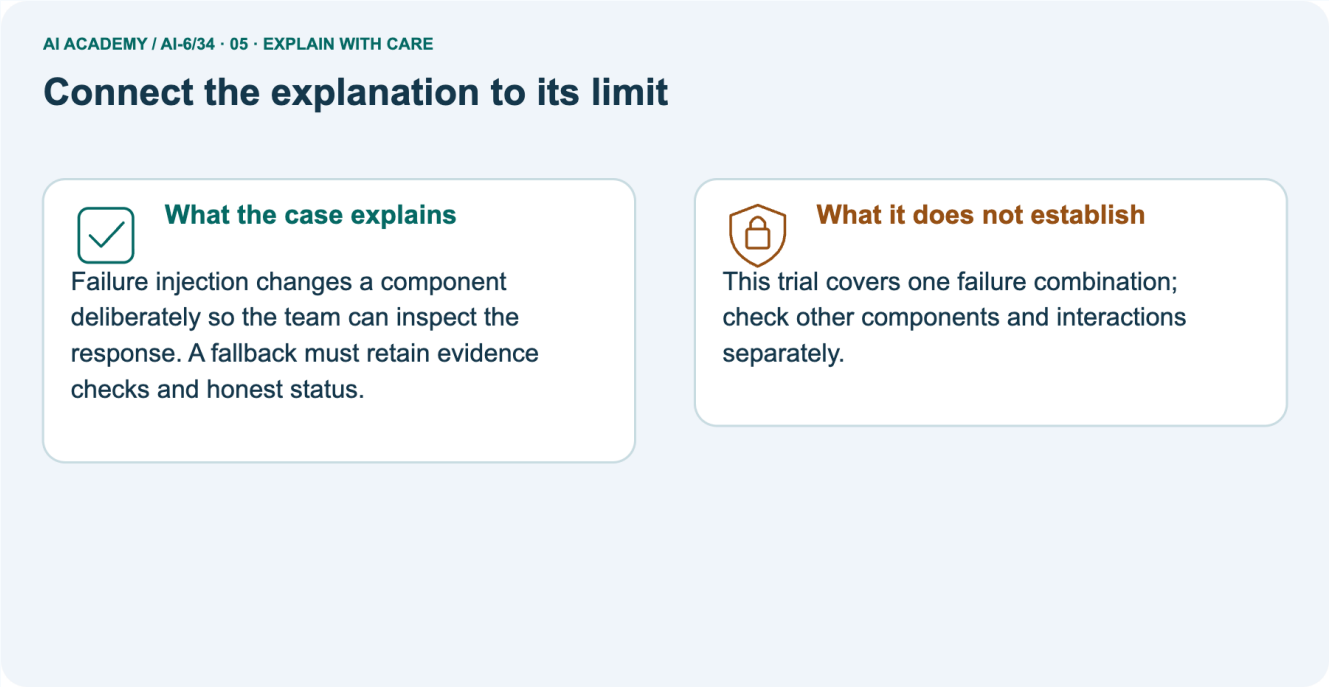
Reason: A template can phrase **evidence** but cannot replace missing **evidence**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A template can phrase **evidence** but cannot replace missing **evidence**.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain **evidence** checks and honest status.

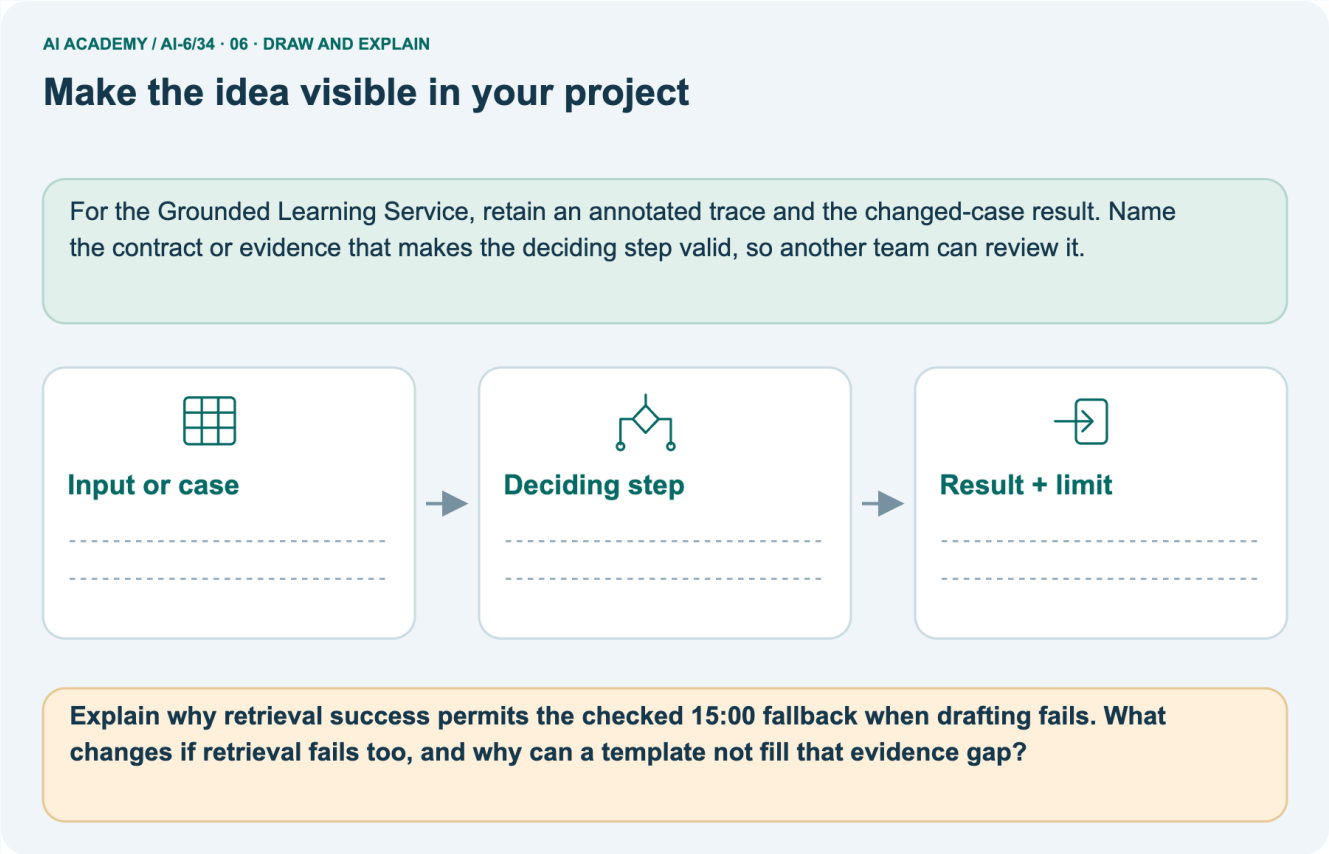
Limit: This trial covers one failure combination; check other components and interactions separately.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This trial covers one failure combination; check other components and interactions separately.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Explain why retrieval success permits the checked 15:00 fallback when drafting fails.

What changes if retrieval fails too, and why can a template not fill that **evidence** gap?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Retrieval supplies F4 and the time 15:00. Disable drafting without changing source **evidence**. Return the checked template citing F4 and record degraded operation.

Changed case: Return unsupported or unavailable; do not use the factual template. A template can phrase **evidence** but cannot replace missing **evidence**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Capstone failure testing: the case

02. Follow the deciding steps

03. Capstone failure testing — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.